

Numerical Analysis — Homework 3

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Deadline: May 8, 2025

1. A function of the form $f(x) = a_0 + a_1\sqrt{x} + a_2x + a_3x^{\frac{3}{2}} + a_4x^2$ satisfies

$$\begin{aligned}f(0) &= 0 \\f(1) &= 1 \\f(4) &= 14 \\f(9) &= 75 \\f(16) &= 244,\end{aligned}$$

Explain how to use divided differences for constructing f . What are a_0, a_1, a_2, a_3 , and a_4 ?

2. Write a computer code that interpolates the function $f(x) = \frac{1}{1+x^2}$ at the points

$$-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$$

by a natural (piecewise cubic) spline p . Also compute the integrals

$$\int_{-5}^5 |f''(x)|^2 dx \quad \text{and} \quad \int_{-5}^5 |p''(x)|^2 dx$$

and interpret your result.